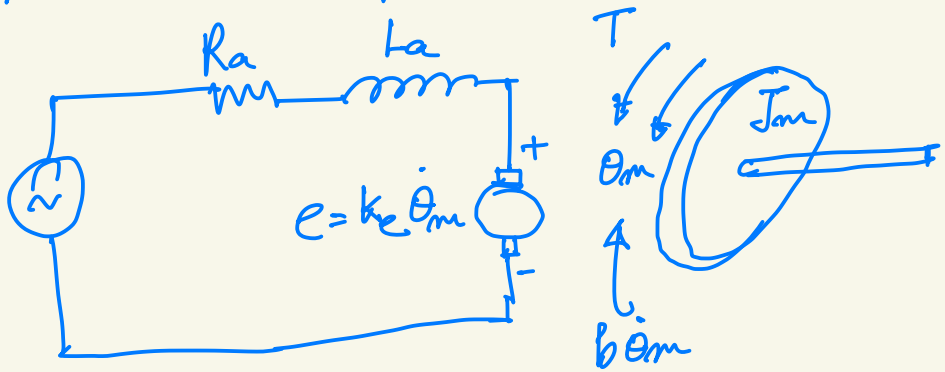




Lecture 27

PID Control of a DC motor:

The circuit model for a DC motor is expressed as below



R_a = Resistance

L_a = inductance

T = Torque

J_m = rotor inertia

The transfer function between the

rotor position θ_m and applied voltage can be found as

$$\frac{\theta_m(s)}{V_a(s)} = G(s) = \frac{k_t}{b[(J_m s + b)(L_a s + R_a) + k_t k_e]}$$

$$G(s) = \frac{k_t}{J_m L_a s^3 + (R_a J_m + b L_a) s^2 + (b R_a + k_t k_e) s}$$

$$J_m = 1.13 \times 10^{-2} \text{ N.m. sec}^2/\text{rad}$$

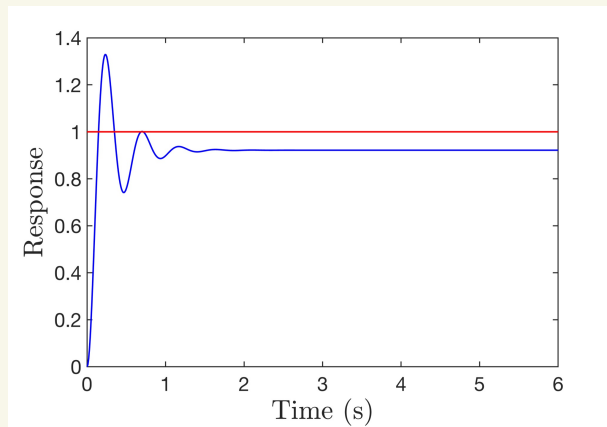
$$b = 0.028 \text{ N.m. sec/rad}$$

$$L_a = 10^{-1} \text{ H} \quad k_t = 0.067 \text{ N.m/amp}$$

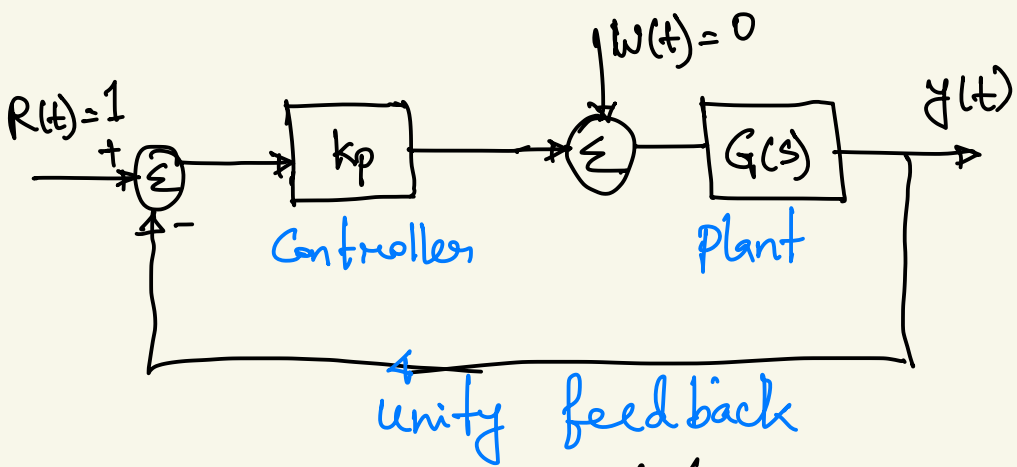
$$R_a = 0.45 \Omega \quad k_e = 0.067 \text{ V.sec/rad}$$

with the above mentioned parameters, let's provide a step

input to the DC motor. In Matlab we perform the calculations and found the following response.



In order to improve the response we try a proportional feedback control.



The combined ^{closed loop} transfer function is as follows with zero noise.

$$Y(s) = \underbrace{\frac{D_{cl}(s) G(s)}{1 + D_{cl}(s) G(s)}}_{T_{cl}(s)} R(s)$$

$$T_{cl}(s) = \frac{D_{cl}(s) G(s)}{1 + D_{cl}(s) G(s)}$$

For proportional control, $D_{cl}(s) = k_p$

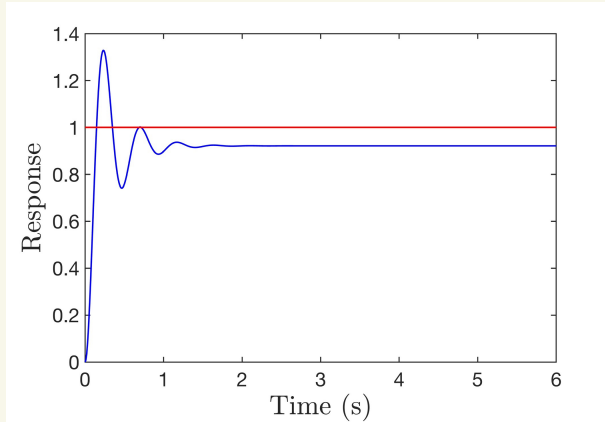
$$k_p \cdot \frac{k_t}{J_m L_a s^3 + (R_a J_m + b \cdot L_a) s^2 + (b R_a + k_t k_e) s}$$

$$T_u(s) = \frac{1 + k_p \cdot \frac{k_t}{J_m L_a s^3 + (R_a J_m + b \cdot L_a) s^2 + (b R_a + k_t k_e) s}}{1 + k_p \cdot \frac{k_t}{J_m L_a s^3 + (R_a J_m + b \cdot L_a) s^2 + (b R_a + k_t k_e) s}}$$

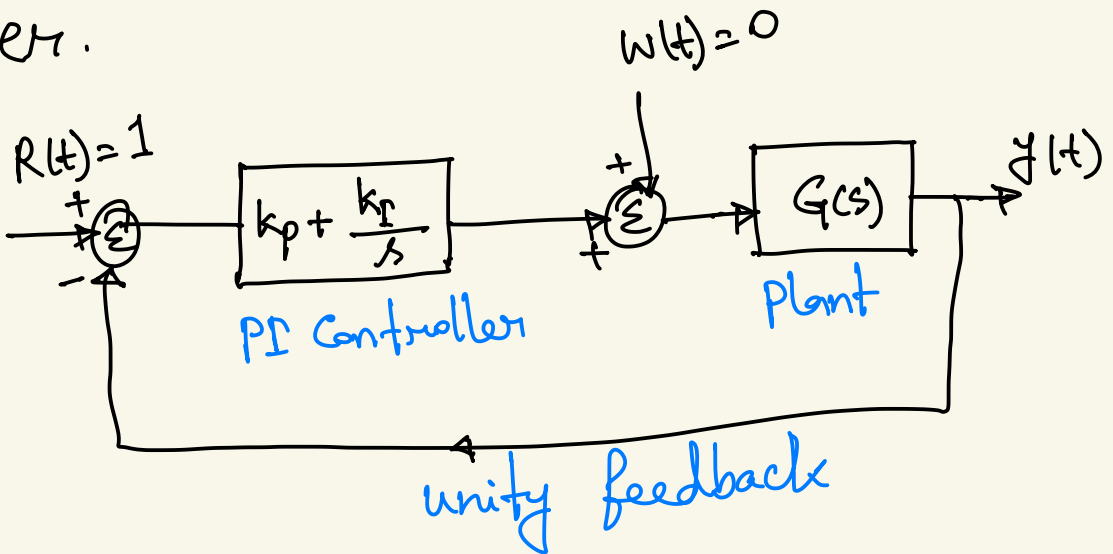
After simplification, we get

$$T_u(s) = \frac{k_p k_t}{J_m L_a s^3 + (R_a J_m + b L_a) s^2 + (b R_a + k_t k_e) s + k_p k_t}$$

From Matlab we note that, the situation does not improve.



let's try a combination of
proportional and integral controll-
er.

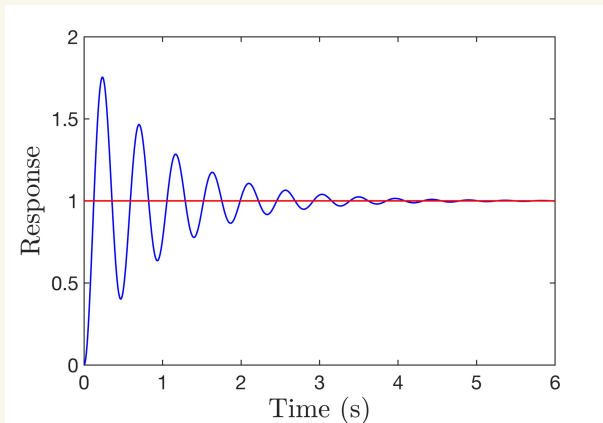


$$D_{cl}(s) = k_p + \frac{k_I}{s}$$

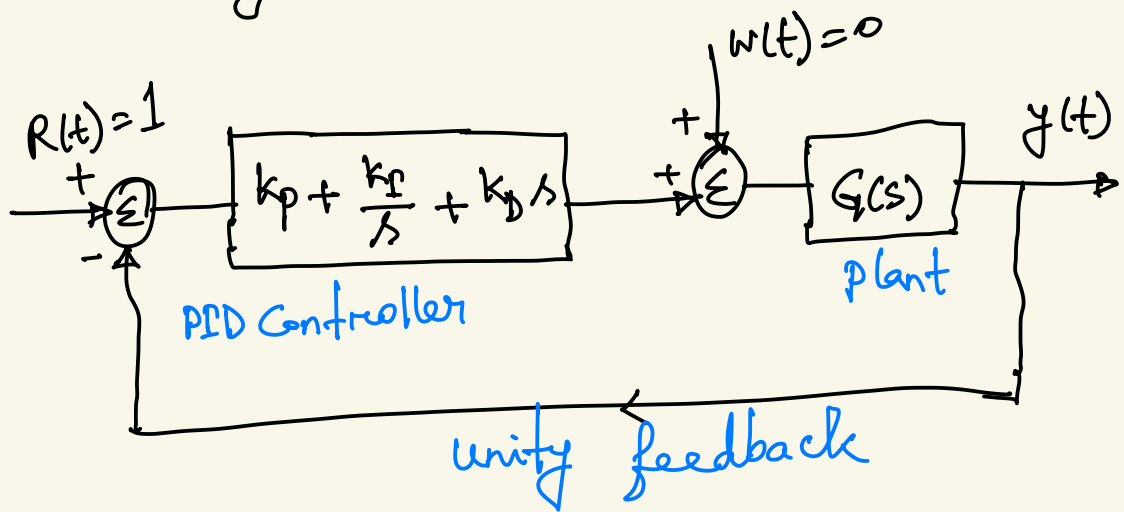
$$T_{cl}(s) = \left(k_p + \frac{k_I}{s}\right) \cdot \frac{k_t}{J_m L s^3 + (R_a J_m + b L a) s^2 + (b R_a + k_t k_e) s}$$

$$1 + \left(k_p + \frac{k_I}{s}\right) \cdot \frac{k_t}{J_m L s^3 + (R_a J_m + b L a) s^2 + (b R_a + k_t k_e) s}$$

$$T_{cl}(s) = \frac{k_p k_t s + k_I k_t}{J_m L s^3 + (R_a J_m + b L a) s^2 + (b R_a + k_t k_e + k_p k_t) s + k_I k_t}$$



In order to decrease the oscillation let's try a PID controller.

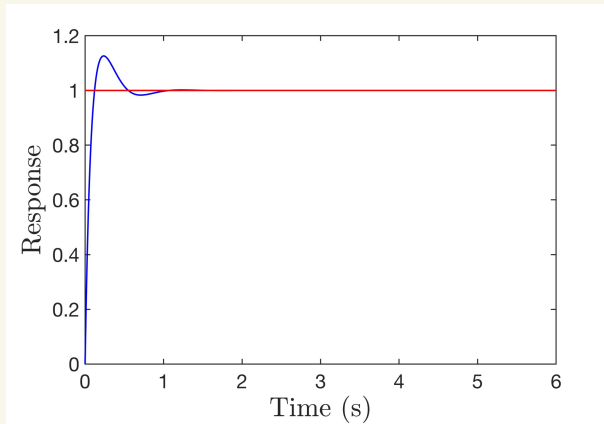


$$D_{cc}(s) = k_p + \frac{k_I}{s} + k_D s$$

$$T_{cc}(s) = \frac{(k_p + \frac{k_I}{s} + k_D s) G(s)}{1 + (k_p + \frac{k_I}{s} + k_D s) G(s)}$$

$$= \frac{(k_D k_t s^2 + k_p k_t s + k_I k_t)}{J_m L a s^3 + (R a J_m + b L a + k_D k_t) s^2 + (b R a + k_t k_t + k_p k_t) s}$$

$$+ k_I k_t$$



With PID controllers we can get the perfect result !